

Ethernet Technology and IEEE 802.3

Ethernet is the dominant data-link technology inside a local area network. The IEEE 802.3 family defines frame format, media access, and physical layers for copper and fiber. Early shared Ethernet used carrier sense multiple access with collision detection, known as CSMA/CD. Hosts listened before transmitting and backed off after a collision.

Modern Ethernet is switched and full duplex, so collisions almost never occur on access ports. A frame still carries a destination MAC address, a source MAC address, an Ethertype, a payload, and a frame check sequence. Switches learn source addresses and forward frames only to the required port, which raises effective bandwidth.

Common physical standards include 10BASE-T, 100BASE-TX, 1000BASE-T, and 10GBASE-T on twisted pair, plus optical variants for backbone links. Auto-negotiation selects speed and duplex. A duplex mismatch between a switch and a workstation produces late collisions, CRC errors, and mysterious slowness.

Jumbo frames can reduce interrupt load on storage networks, but every device on the path must agree on the maximum transmission unit. Administrators therefore treat jumbo frames as an optional optimization, not a default.

Understanding Ethernet is essential for LAN design. Cabling categories, patch quality, and correct speed settings determine whether the network behaves like a reliable campus fabric or like a collision-prone shared medium from the 1990s.